



CAESAR II Users Guide

Hexagon Documentation

Generated 03/27/2026

Table of Contents

[CAESAR II Neutral File](#)

[Version and Job Title Information](#)

[Control Information](#)

[Basic Element Data](#)

[Auxiliary Element Data](#)

[Miscellaneous Data Group #1](#)

[Units Conversion Data](#)

[Nodal Coordinate Data](#)

[Neutral File Generator](#)

CAESAR II Neutral File

Main window Interfaces tab: **Generic > CAESAR II Neutral File**



Converts a CAESAR II input file to a neutral file or converts a neutral file to a CAESAR II input file.

The CAESAR II neutral file, an ASCII text file with the .cii extension, enables access to any particular data item from an _A input file, to enable a complete _A file to be built from a CAD application. The .cii file can be used to send data either to or from the _A file. The name of the file used or generated by this interface is the CAESAR II jobname.

NOTE The content and format described in this section is subject to change as a function of the enhancements made to CAESAR II. Every effort is being made to keep "drastic" changes to a minimum.

Several third-party CAD applications, such as AVEVA[®]'s PDMS[®] and Jacobus' PlantSpace[™], also support this neutral file. If you prefer, instead of launching this interface from **Tools > External Interface**, the processor can be run in "batch mode" from either a batch file or the command line as shown below.

```
f:\ProgramDirectory\iecho f:\DataDirectory\NeutralFile.cii
```

where:

f:\ProgramDirectory and f:\DataDirectory must be changed as appropriate.

The .CII file, is divided into sections which organize the piping data in logical groupings. Section divisions are denoted in the neutral file by the '\$#' character sequence found in columns 1 and 2. The token following the '\$#' character sequence is a section identifier that is used by the software for data sequencing purposes, and to aid you in reading the neutral file. For each item listed on the following pages, the necessary FORTRAN format for the input/output is provided. The variables listed below are used in dimensioning arrays.

- N1—Base memory allocation quantity used to set array sizes. For example, if N1=2,000, your neutral file can handle up to 2,000 elements.
- N2—1/2 N1
- N3—1/3 N1
- N4—1/4 N1

- N5—1/5 N1
- N6—N1/13.33

 **NOTE** For information on editing ASCII text, see ASCII Text File Rules.

Version and Job Title Information

#\$ VERSION

Provides section header information.

Use FORTRAN format (2X, 2G13.6, I8) to write the values of the following variables on the first line of the neutral file:

- GVERSION is the version of the neutral file interface being used. This corresponds to the major version number of CAESAR II (4 for 4.x., for example).
- RVERSION is the specific CAESAR II version generating this file, for example 4.50.
- ICODEPAGE is the Windows (ANSI) code page identifier, for example 1252.

The next 60 lines of 75 characters each are reserved for the title-page text. Use FORTRAN format (2X, A75).

The last line of the job title array, if blank, is set by this transfer interface. The text that is set here indicates that the file was created by the CAESAR II interface.

Control Information

#\$ CONTROL

Provides the section division header. The **#\$** and space are required, and the word **CONTROL** is in all uppercase.

Use the FORTRAN format (2X, 6I13) to write the values of the following variables on the next line of the neutral file:

NUMELT

Defines the number of piping elements (spreadsheets) in the input file.

NUMNOZ

Defines the number of nozzles in the input file.

NOHGRS

Defines the number of spring hangers in the input file.

NONAM

Defines the number of Node Name data blocks in the input file.

NORED

Defines the number of reducers in the input file.

NUMFLG

Defines the number of flanges in the input file.

Write 13 items that contain the number of auxiliary data types used in the input file followed by the vertical axis indicator. Use the FORTRAN format (2X, 6I13). These 13 values are:

- The number of bend auxiliary data blocks in the input file.
- The number of rigid-element auxiliary data blocks in the input file.
- The number of expansion-joint auxiliary data blocks in the input file.
- The number of restraint auxiliary data blocks in the input file.
- The number of displacement auxiliary data blocks in the input file.
- The number of force/moment auxiliary data blocks in the input file.
- The number of uniform-load auxiliary data blocks in the input file.
- The number of wind-load auxiliary data blocks in the input file.
- The number of element-offset auxiliary data blocks in the input file.
- The number of allowable-stress auxiliary data blocks in the input file.
- The number of intersection auxiliary data blocks in the input file. IZUP flag. Equal to zero (0) for the global -Y axis vertical; equal to 1 for the global -Z axis vertical.
- The number of (nozzle) equipment limits data blocks in the input file.

Basic Element Data

NOTE Where the FORTRAN format is (2X, 6G13.6), you can change the format to (2X, 6(1p,E13.6)) to write the value with seven significant digits.

#\$ ELEMENTS - Contains integer and real data for each element in the input file. The data is organized as shown below.

1. *Real* values for element "i"
2. *Integer* values for element "i"
3. *Real* values for element "i+1"
4. *Integer* values for element "i+1"

These *real* and *integer* values are stored in arrays that are described below.

A 98-member array (REL) contains the real basic-element data. The REL array is dimensioned (N1,98).

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 53 items on the appropriate nine lines of the neutral file.

1. FROM node number
2. TO node number
3. Delta X
4. Delta Y
5. Delta Z
6. Diameter (value stored here is actual OD)
7. Wall Thickness (actual)
8. Insulation Thickness
9. Corrosion Allowance
10. Thermal Expansion Coefficient #1 (or Temperature #1)
11. Thermal Expansion Coefficient #2 (or Temperature #2)
12. Thermal Expansion Coefficient #3 (or Temperature #3)

13. Thermal Expansion Coefficient #4 (or Temperature #4)
14. Thermal Expansion Coefficient #5 (or Temperature #5)
15. Thermal Expansion Coefficient #6 (or Temperature #6)
16. Thermal Expansion Coefficient #7 (or Temperature #7)
17. Thermal Expansion Coefficient #8 (or Temperature #8)
18. Thermal Expansion Coefficient #9 (or Temperature #9)
19. Pressure #1
20. Pressure #2
21. Pressure #3
22. Pressure #4
23. Pressure #5
24. Pressure #6
25. Pressure #7
26. Pressure #8
27. Pressure #9
28. Elastic Modulus (cold)
29. Poisson's Ratio
30. Pipe Density
31. Insulation Density
32. Fluid Density
33. Minus Mill Tolerance
34. Plus Mill Tolerance
35. Seam Weld (1=Yes, 0=No)
36. Hydro Pressure

- 37. Elastic Modulus (Hot #1)
- 38. Elastic Modulus (Hot #2)
- 39. Elastic Modulus (Hot #3)
- 40. Elastic Modulus (Hot #4)
- 41. Elastic Modulus (Hot #5)
- 42. Elastic Modulus (Hot #6)
- 43. Elastic Modulus (Hot #7)
- 44. Elastic Modulus (Hot #8)
- 45. Elastic Modulus (Hot #9)
- 46. "wL" Factor
- 47. Element Orientation Angle (To End), FEATools Data Only
- 48. Element Orientation Angle (From End), FEATools Data Only
- 49. Cladding Thickness
- 50. Cladding Density
- 51. Insulation + Cladding Weight/length
- 52. Refractory Thickness
- 53. Refractory Density

Non-specified *real* values are assigned a value of 0.0 by this interface.

If the delta coordinates are not specified, they default to zero. If the **To/From** fields are not specified, it is considered an error.

Use the FORTRAN format (7X, I5, 1X, A500) to write the values of the following items on the next lines.

 **NOTE** The value before the string indicates the length of the string.

- 1. Element Name
- 2. Line Number

3. Line Color and Line Visibility, using the format (2X, 6G13.6)

An 18-member array (IEL) contains the pointers to the auxiliary data arrays. The IEL array is dimensioned (N1,18).

NOTE At this time, only 15 of the members of this array are used.

Use the FORTRAN format (2X, 6I13) to write the values of the following 15 items on the next three lines of the neutral file.

1. Pointer to **Bend Auxiliary** field. This indicates where in the bend auxiliary array the bend data for the current element can be found.
2. Pointer to **Rigid Element Auxiliary** field.
3. Pointer to **Expansion Joint Auxiliary** field.
4. Pointer to **Restraint Auxiliary** field.
5. Pointer to **Displacement Auxiliary** field.
6. Pointer to **Force/Moment Auxiliary** field.
7. Pointer to **Uniform Load Auxiliary** field.
8. Pointer to **Wind Load Auxiliary** field.
9. Pointer to **Element Offset Auxiliary** field.
10. Pointer to **Allowable Stress Auxiliary** field.
11. Pointer to **Intersection Auxiliary** field.
12. Pointer to **Node Name Auxiliary** field.
13. Pointer to **Reducer Auxiliary** field.
14. Pointer to **Flange Auxiliary** field.
15. Pointer to **Nozzle/Equipment Check Auxiliary** field.

NOTE When there is no auxiliary data of a particular type associated with the current element use a pointer value of zero.

Auxiliary Element Data

NOTE Where the FORTRAN format is (2X, 6G13.6), you can change the format to (2X, 6(1p,E13.6)) to write the value with seven significant digits.

#\$ AUX_DATA

Contains the auxiliary data corresponding to the elements. This data is arranged in the same order as the IAUXAU array described previously. For example, if IAUXAU(1) contains a 3, then there are three bends in the model, and their data is found next in the neutral file. Likewise, if IAUXAU(2) contains a 5, then there are five rigid elements in the model and their data follows the bend data.

Each set of auxiliary data is separated by a sub-section header. If a particular value in IAUXAU is zero, then only the subsection header is written to the neutral file.

The data storage for these arrays is allocated at runtime based on the available free system memory. These arrays are allocated proportionally as a percentage of the n-number of elements allowed. Four proportions are used: 1/2, 1/3, 1/4, and 1/5. These proportions correspond to the variables: N2, N3, N4, and N5. Maintain these proportions to ensure that the neutral file reader can accept the file.

#\$ NODENAME

Defines the Node Name data. To maintain downward compatibility, this section is optional. The data for each element set of node names in the input file is listed here.

A two-member array (NAM) defines each set of node names. The NAM array is dimensioned (N6, 2). Use the FORTRAN format (2X, A10, 16X, A10) to read the character name of the **FROM** node and then that of the **TO** node.

#\$ BEND

Defines the bend data. The data for each bend in the input file is listed here.

A 15-member array (BND) defines each bend. The BND array is dimensioned (N3,15).

Only 13 items are currently used. Use the FORTRAN format (2X, 6G13.6) to write the values of the following 13 items on the next three lines of the neutral file.

1. Bend radius
2. Type: 1 - single flange; 2 - double flange; 0 or blank - welded
3. Angle to node position #1

4. Node number at position #1
5. Angle to node position #2
6. Node number at position #2
7. Angle to node position #3
8. Node number at position #3
9. Number of miter cuts
10. Fitting thickness of bend if different from the pipe
11. Seam Weld (1=Yes, 0=No)
12. Bend flexibility (K) factor
13. Weld strength reduction factor WL
14. Overlay Thickness

#\$ RIGID

Defines the rigid data. The data for each rigid in the input file is listed here.

A two-element array (RIG) is used for each rigid. The RIG array is dimensioned (N3,2). The elements of the array represent:

- Rigid Weight
- Rigid Type

Use the FORTRAN format (2X, 6G13.6) to write the value.

#\$ EXPJT

Defines the expansion joint data. The data for each expansion joint in the input file is listed here. The EXP array is dimensioned (N5,5).

Use FORTRAN format (2X, 6G13.6) to write the values of the following five items on the next line of the neutral file.

1. Axial stiffness
2. Transverse stiffness

3. Bending stiffness
4. Torsional stiffness
5. Effective inside bellows diameter

#\$ RESTRAINT

Defines the restraint data.

The data for each restraint auxiliary data block in the input file is listed here. The RES array is dimensioned (N2,102).

Each restraint degree of freedom consists of four lines: nine values on two lines and then tag and GUID text on two lines. The four lines are repeated six times for each restraint, one for each DOF.

Lines 1 and 2: Degrees of Freedom

Use the FORTRAN format (2X, 6G13.6) to write the values of the following items on the first two lines.

1. Restraint node number
2. Restraint type

The restraint type is an integer value whose valid range is from 1 to 62.

1 - 5	ANC	X	Y	Z	RX
6 - 10	RY	RZ	GUI	LIM	XSNB
11 - 15	YSNB	ZSNB	+X	+Y	+Z
16 - 20	-X	-Y	-Z	+RX	+RY
21 - 25	+RZ	-RX	-RY	-RZ	+LIM
26 - 30	-LIM	XROD	YROD	ZROD	+XROD
31 - 35	+YROD	+ZROD	-XROD	-YROD	-ZROD
36 - 40	X2	Y2	Z2	RX2	RY2
41 - 45	RZ2	+X2	+Y2	+Z2	-X2
46 - 50	-Y2	-Z2	+RX2	+RY2	+RZ2
51 - 55	-RX2	-RY2	-RZ2	XSPR	YSPR
56 - 60	ZSPR	+XSNB	+YSNB	+ZSNB	-XSNB
61 - 62	-YSNB	-ZSNB			

3. Restraint stiffness
4. Restraint gap
5. Restraint friction coefficient
6. Restraint connecting node

7. X direction cosine
8. Y direction cosine
9. Z direction cosine

Lines 3 and 4: Text

Use the FORTRAN format (7X, I5, 1X, A100) to write the values of the following two items on the next two lines of the neutral file.

1. Support Tag
2. Support GUID

NOTE The value at the beginning of each string indicates the length of the string. If the string is blank, the value is 0.

#\$ DISPLMNT

Defines the displacement data. The data for each displacement auxiliary data block in the input file is listed here.

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 55 items on the next lines of the neutral file. The DIS array is dimensioned (N3,110).

This requires 10 lines in the neutral file for each displacement specification. This means 20 lines total for each displacement auxiliary.

	1) first displacement node number					
{vector 1}	2) DX	3) DY	4) DZ	5) RX	6) RY	7) RZ
{vector 2}	8) DX	9) DY	10) DZ	11) RX	12) RY	13) RZ
{vector 3}	14) DX	15) DY	16) DZ	17) RX	18) RY	19) RZ
{vector 4}	20) DX	21) DY	22) DZ	23) RX	24) RY	25) RZ
{vector 5}	26) DX	27) DY	28) DZ	29) RX	30) RY	31) RZ
{vector 6}	32) DX	33) DY	34) DZ	35) RX	36) RY	37) RZ
{vector 7}	38) DX	39) DY	40) DZ	41) RX	42) RY	43) RZ
{vector 8}	44) DX	45) DY	46) DZ	47) RX	48) RY	49) RZ
{vector 9}	50) DX	51) DY	52) DZ	53) RX	54) RY	55) RZ
	56) second displacement node number					
{vector 1}	57) DX	58) DY	59) DZ	60) RX	61) RY	62) RZ
{vector 2}	63) DX	64) DY	65) DZ	66) RX	67) RY	68) RZ
{vector 3}	69) DX	70) DY	71) DZ	72) RX	73) RY	74) RZ
{vector 4}	75) DX	76) DY	77) DZ	78) RX	79) RY	80) RZ
{vector 5}	81) DX	82) DY	83) DZ	84) RX	85) RY	86) RZ
{vector 6}	87) DX	88) DY	89) DZ	90) RX	91) RY	92) RZ
{vector 7}	93) DX	94) DY	95) DZ	96) RX	97) RY	98) RZ
{vector 8}	99) DX	100) DY	101) DZ	102) RX	103) RY	104) RZ
{vector 9}	105) DX	106) DY	107) DZ	108) RX	109) RY	110) RZ

These 55 items are repeated twice for the two possible displacements defined on the auxiliary.

NOTE Unspecified displacement values (free-displacement degrees of freedom, for example) are designated by using a value of 9999.99.

#\$ FORCMNT

Defines the start of the force/moment data. The data for each force/moment auxiliary data block in the input file is listed here.

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 55 items on the next ten lines of the neutral file. The FOR array is dimensioned (N3,110).

This requires ten lines in the neutral file for each force/moment specification. This means 20 lines total for each force/moment auxiliary data block.

	1) first force/moment node number					
{vector 1}	2) FX	3) FY	4) FZ	5) MX	6) MY	7) MZ
{vector 2}	8) FX	9) FY	10) FZ	11) MX	12) MY	13) MZ
{vector 3}	14) FX	15) FY	16) FZ	17) MX	18) MY	19) MZ
{vector 4}	20) FX	21) FY	22) FZ	23) MX	24) MY	25) MZ
{vector 5}	26) FX	27) FY	28) FZ	29) MX	30) MY	31) MZ
{vector 6}	32) FX	33) FY	34) FZ	35) MX	36) MY	37) MZ
{vector 7}	38) FX	39) FY	40) FZ	41) MX	42) MY	43) MZ
{vector 8}	44) FX	45) FY	46) FZ	47) MX	48) MY	49) MZ
{vector 9}	50) FX	51) FY	52) FZ	53) MX	54) MY	55) MZ
	56) second displacement node number					
{vector 1}	57) FX	58) FY	59) FZ	60) MX	61) MY	62) MZ
{vector 2}	63) FX	64) FY	65) FZ	66) MX	67) MY	68) MZ
{vector 3}	69) FX	70) FY	71) FZ	72) MX	73) MY	74) MZ
{vector 4}	75) FX	76) FY	77) FZ	78) MX	79) MY	80) MZ
{vector 5}	81) FX	82) FY	83) FZ	84) MX	85) MY	86) MZ
{vector 6}	87) FX	88) FY	89) FZ	90) MX	91) MY	92) MZ
{vector 7}	93) FX	94) FY	95) FZ	96) MX	97) MY	98) MZ
{vector 8}	99) FX	100) FY	101) FZ	102) MX	103) MY	104) MZ
{vector 9}	105) FX	106) FY	107) FZ	108) MX	109) MY	110) MZ

#\$ UNIFORM

Defines the start of the uniform load data. The data for each uniform load in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following 36 items on the next six lines of the neutral file. The UNI array is dimensioned (N5,36). Nine vectors of four values each (three directions and a G-load flag) are used.

This requires six lines in the neutral file for each uniform load auxiliary data block.

G-flag is 1 for the input values in G's and 0 for input values in force-per-length notation.

vector 1 & 2:	UX1	UY1	UZ1	G-flag1	UX2	UY2
vector 2 & 3:	UZ2	G-flag2	UX3	UY3	UZ3	G-flag3
vector 4 & 5:	UX4	UY4	UZ4	G-flag4	UX5	UY5
vector 5 & 6:	UZ5	G-flag5	UX6	UY6	UZ6	G-flag6
vector 7 & 8:	UX7	UY7	UZ7	G-flag7	UX8	UY8
vector 8 & 9:	UZ8	G-flag8	UX9	UY9	UZ9	G-flag9

#\$ WIND

Defines the start of the wind/wave data. The data for each wind/wave specification in the input file is listed here. The WIND array is dimensioned (N5,6).

Use the FORTRAN format (2X, 6G13.6) to write the set of values on the next line of the neutral file. This requires a single line in the neutral file for each wind auxiliary.

The data items on each line are as follows:

1. Entry type (0.0 for Wind, 1.0 for Wave, 2.0 for Off)
2. Wind shape factor or wave drag coefficient
3. Wave added mass coefficient
4. Wave lift coefficient
5. Wave marine growth
6. Marine growth density

#\$ OFFSETS

Defines the start of the element offset data. The data for each offset pipe in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following six items on the next line of the neutral file. The OFF array is dimensioned (N5,6).

This requires a single line in the neutral file for each offset auxiliary.

1. Element FROM node offset in X direction
2. Element FROM node offset in Y direction
3. Element FROM node offset in Z direction
4. Element TO node offset in X direction
5. Element TO node offset in Y direction
6. Element TO node offset in Z direction

#\$ ALLOWBLS

Defines the start of the allowable stress data. The data for each allowable spec in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following 168 items on the next 28 lines of the neutral file. The array is dimensioned (N5,168).

1. Cold allowable stress
2. Hot allowable for thermal case #1
3. Hot allowable for thermal case #2
4. Hot allowable for thermal case #3
5. Code cyclic stress range reduction factor for thermal case #1
6. Code cyclic stress range reduction factor for thermal case #2
7. Code cyclic stress range reduction factor for thermal case #3
8. Eff.
9. Sy

10. Fac
11. Pmax
12. Piping code id
13. Hot allowable for thermal case #4
14. Hot allowable for thermal case #5
15. Hot allowable for thermal case #6
16. Hot allowable for thermal case #7
17. Hot allowable for thermal case #8
18. Hot allowable for thermal case #9
19. Code cyclic stress range reduction factor for thermal case #4
20. Code cyclic stress range reduction factor for thermal case #5
21. Code cyclic stress range reduction factor for thermal case #6
22. Code cyclic stress range reduction factor for thermal case #7
23. Code cyclic stress range reduction factor for thermal case #8
24. Code cyclic stress range reduction factor for thermal case #9

Items 25 through 32 represent Cycles, and items 33 through 40 represent Stresses for eight BW (butt-weld)/Class D Fatigue pairs.

Items 41 through 48 represent Cycles, and items 49 through 56 represent Stresses for eight FW (fillet-weld)/Class E Fatigue pairs.

Items 57 through 64 represent Cycles, and items 65 through 72 represent Stresses for eight Class F Fatigue pairs used with TD/12 piping code.

Items 73 through 80 represent Cycles, and items 81 through 88 represent Stresses for eight Class G Fatigue pairs used with TD/12 piping code.

Items 89 through 96 represent Cycles, and items 97 through 104 represent Stresses for eight Class W Fatigue pairs used with TD/12 piping code.

Item 105 – Elastic Modulus correction

Item 106 – has different meanings based on the active piping code: Allowed Cycles Maximum (per B31.3); Material Composition/Type (per HPGSL and JPI).

Item 107 – UTS ambient.

Item 108 – Code specific switch.

Items 109 through 117 represent nine SY values at temperature.

Items 118 through 126 represent nine UTS values at temperature.

B31.12 (items 127-129):

Item 127 – Material performance factor, M_f or H_f .

Item 128 – Material coefficient, Y .

Item 129 – Design factor, F

ISO 14692-2017 (items 130-136) – A_0 (130), A_2 , Buried check box, E_{hb} , Shape factor D_f , vertical deflection y/D (135), Curve radius R (136).

DNV-2017 (items 137-149)- α_h (137), α_{fab} , Supplementary Req, Material LP, Burst Ope, Burst Test, Collapse, Prop Buckling, LCC, DCC, γ_c , R , α_{gw} .

Items 150 through 156 are currently unused. Write the value of 0.000000.

Items 157 through 165 – Temperature derating factors for thermal cases #1 thru #9.

Item 166 – B31.12 expansion allowable switch .

Items 167 and 168 are currently unused.

NOTES

- Some of these items (notably 8-24) may have different meanings based on the active piping code.
- Piping code ISO-14692 has special mapping for the first 24 items.

#\$ SIF&TEES

Defines the start of the SIF/TEE data.

Use FORTRAN format (2X, 6G13.6) to write the input values for each of the two tees nodes specified. The SIF array is dimensioned (N4,84).

The neutral file requires seven lines for each SIF/TEE element. Each line has six values. Unspecified or unused values are filled with 0.0. Each element has two tee nodes, so specify 14 lines total for each SIF/TEE element.

1. Node - The intersection node number.
2. Type - The intersection type code. If not specified this auxiliary data is only used to specify SIFs.
3. In-Plane SIF, i_i .
4. Out-Plane SIF, i_o .
5. Weld, d - Circumferential weld mismatch, used for butt welds and tapered transitions.
6. Fillet - Fillet leg length.
7. Pad Thk - Thickness of the reinforcing pad.
8. Ftg Ro - Fitting outside radius for branch connections.
9. Crotch R - Crotch radius of the formed lip on an extruded welding tee.
10. Weld ID - Weld ID value.
11. B1 - Code-specific value.
12. B2 - Code-specific value.
13. In-Plane Index, I_i .
14. Out-Plane Index (I_o).
15. Axial SIF, i_a .
16. Torsion SIF (i_t).
17. Pressure SIF, i_p .
18. Pressure Index, I_p .
19. Not used.
20. Not used.
21. Not used.

22. Not used.

23. Code-related note options.

When the check box is selected, displays 1. When the check box is cleared, displays 0. When the check box is not specified, displays 9999.99.

24. Code-related note options.

When the check box is selected, displays 1. When the check box is cleared, displays 0. When the check box is not specified, displays 9999.99.

25. Axial Index, I_a .

26. Torsion Index (I_t).

27. Surface Node (B31J).

28. Not used (B31J).

29. Not used (B31J).

30. Not used B31J).

31. Number of Flanges, $N_{o. \text{ Flanges}}$ (B31J).

32. Mismatch - Maximum (B31J).

33. Reinf. Type (B31J).

34. Radius r_2 (B31J).

35. Reinforcement Thickness t_n (B31J).

36. Reinforcement Length L_1 (B31J).

37. Angle θ_n (B31J).

38. Taper Large End y (B31J).

39. Effective Branch Thickness t . (B31J).

40. Not used (B31J).

41. Not used (B31J).

42. Not used (B31J).

These values repeat for the second tee node.

NOTE Some values may have different meanings based on your selected the piping code. For piping input specifics, see SIFs & Tees.

#\$ REDUCERS

Defines the start of the REDUCER data.

Use FORTRAN format (2X, 6G13.6) to write the input values. The RED array is dimensioned (N6,10).

The neutral file requires two lines in for each REDUCER specified. Each line has 5 values. Unspecified or unused values are filled with 0.0.

1. Diameter 2 - Diameter at the **To** node of the reducer element.
2. Thickness 2 - Wall thickness at the **To** node of the reducer element.
3. Alpha - Slope of the reducer transition.
4. R1 - Transition radius for the large end of the reducer (TD/12 piping codes).
5. R2 - Transition radius for the small end of the reducer (TD/12 piping codes).
6. Not used.
7. L2 - Length of the cylindrical portion at the small end of the reducer (B31J).
8. Not used.
9. Not used.
10. Not used.

#\$ FLANGES

Defines the FLANGE data. The data for each FLANGE spec in the input file is listed here. There are 72 data values used to describe a flange.

1. FROM/TO (0 = FROM, 1 = TO, 2= BOTH)
2. METHOD (0 = PEQ, 1 = ASME NC)
3. GASKET OR BOLT CIRCLE DIAMETER, DEPENDING ON METHOD
4. BOLT AREA (ASME METHOD ONLY)
5. SYNC (ASME METHOD ONLY)

6. SY1 (ASME METHOD ONLY)
7. SY2 (ASME METHOD ONLY)
8. SY3 (ASME METHOD ONLY)
9. SY4 (ASME METHOD ONLY)
10. SY5 (ASME METHOD ONLY)
11. SY6 (ASME METHOD ONLY)
12. SY7 (ASME METHOD ONLY)
13. SY8 (ASME METHOD ONLY)
14. SY9 (ASME METHOD ONLY)
15. 15-24 CLASS NAME, (40 CHAR MAX)
16. 25-48 24 TEMPERATURES OF THE TEMP/PRESS RATING CURVE (PEQ METHOD ONLY)
17. 49-72 24 PRESSURES OF THE TEMP/PRESS RATING CURVE (PEQ METHOD ONLY)

These values are arranged in the neutral file on 12 lines using a format of (2X, 6G13.6) unless otherwise specified:

Line 1: Flange items 1-5

Line 2: Flange items 6-11

Line 3: Flange items 12-14

Line 4: Class Name, using a format of (2X, A40)

Line 5: Flange items 25-30

Line 6: Flange items 31-36

Line 7: Flange items 37-42

Line 8: Flange items 43-48

Line 9: Flange items 49-54

Line 10: Flange items 55-60

Line 11: Flange items 61-66

Line 12: Flange items 67-72

All 12 lines must be written to the neutral file for each flange. Unused fields/values can be represented by 0.00.

#\$ EQUIPMNT

Defines the Equipment/Nozzle Check data. The data for each EQUIPMNT spec in the input file is listed here. There are two sets of 17 data values that use a format of (2X, 6G13.6) as shown below:

1. Node Number
2. Limiting load value FX
3. Limiting load value FY
4. Limiting load value FZ
5. Limiting load value MX
6. Limiting load value MY
7. Limiting load value MZ
8. Reference axis direction cosine CosX
9. Reference axis direction cosine CosY
10. Reference axis direction cosine CosZ
11. Flange rating
12. Interaction method: 0=absolute; 1=SRSS; 2=Unity Check;

Items 13-17 are spares represented by 0.00. These values are arranged in the neutral file on six lines. All six lines must be written to the neutral file for each **Nozzle/Equipment** check. Unused fields/values can be represented by 0.00.

Miscellaneous Data Group #1

NOTE Where the FORTRAN format is (2X, 6G13.6), you can change the format to (2X, 6(1p,E13.6)) to write the value with seven significant digits.

#\$ MISCEL_1

Contains the material ID (RRMAT) for each element in the input file, the nozzle data (VFLEX), the hanger data, and the execution options.

Material ID

Contains the material ID number in the first array for each element in the input file. Use the FORTRAN format (2X, 6G13.6).

The RRMAT array is dimensioned (N1). The material IDs range from 1 to 699 (See the User's Guide for details).

The number of lines required to write the RRMAT array in the neutral file is determined by the following FORTRAN routine:

```
NLINES = NUMELT / 6
```

```
IF(MOD(NUMELT,6).NE.0)THEN  
NLINES = NLINES + 1
```

```
ENDIF
```

Nozzles

Describes the flexible WRC-297, PD 5500, API 650, and Custom nozzles in the input file.

NOTE The value 9999.99 represents infinity or an undefined term.

Use the FORTRAN format (2X, 6G13.6). The nozzle (VFLEX) contains 22 values for each nozzle in the input. This requires four lines in the neutral file.

For WRC-297 nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (0 = WRC-297)
4. Nozzle outside diameter (in.)

5. Nozzle wall thickness (in.)
6. Vessel outside diameter (in.)
7. Vessel wall thickness (in.)
8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Dist. to stiffeners or head (in.)
11. Dist. to opposite side stiffeners or head (in.)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Vessel temperature (optional) (°F)
16. Vessel material # (optional)
17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For API 650 nozzles, the items are:

1. Nozzle node number
2. Tank node number (optional)
3. Nozzle type indicator (1.0 = API-650)
4. Nozzle outside diameter (in.)
5. Nozzle wall thickness (in.)

6. Tank outer diameter (in.)
7. Tank wall thickness (in.)
8. Spare (not used)
9. Reinforcing on shell (1) or nozzle (2)
10. Nozzle height (in.)
11. Fluid height (in.)
12. Displacement vector (integer value between 1 and 9, 0 if unused)
13. Specific gravity of fluid
14. Tank thermal expansion coefficient (in/in/deg)
15. Delta temperature (°F)
16. Elastic modulus (psi)
17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For PD 5500 nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (.0 = PD 5500)
4. Nozzle outside diameter (in.)
5. Vessel type (0-cylinder, 1-sphere)
6. Vessel outside diameter (in.)

7. Vessel wall thickness (in.)
8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Dist. to stiffeners or head (in.)
11. Dist. to opposite side stiffeners or head (in.)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Vessel Temperature (optional) (°F)
16. Vessel Material # (optional)
17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For Custom nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (3.0 = Custom)
4. Nozzle outside diameter (in.)
5. Nozzle wall thickness (in.)
6. Vessel outside diameter (in.)
7. Vessel wall thickness (in.)

8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Spare (not used)
11. Spare (not used)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Spare (not used)
16. Vessel material # (optional)
17. Spare (not used)
18. Spare (not used)
19. Axial stiffness (lb.\in.)
20. In-plane stiffness (in.-lb./deg.)
21. Out-of-plane stiffness (in.-lb./deg.)
22. Torsional stiffness (in.-lb./deg.)

Hangers

Describes the spring hangers in the input file. Some of the hanger data listed below represents uninitialized data. In the instances where this uninitialized data represent infinite values (such as maximum travel limit and available space), it is reported here as 9999.99.

The next line contains values for the following parameters in FORTRAN format (2X, I13, 5G13.6):

- IDFTABLE is the default hanger table.
- DEFVAR is the default for allowed load variation.
- DEFRIG is the default for rigid support displacement criteria.
- DEFMXTRAVEL is the default for maximum allowed travel.

- DEFSHTSPR is the default for allowing short range springs (0=no 1=yes).
- DEFMUL is the default multi-load case design option.

The next line contains values for the following parameters in the FORTRAN format (2X, 5I13):

- IDFOPER is the default number of hanger design operating cases (always 1).
- IACTCLD is the default cold load calculation switch (0=no, 1=yes).
- IHGRLDS is the number of hanger operating loads (0 -3).
- IACTUAL is the load case defining actual cold loads.
- IMULTIOPTS is the multi-load case design option (1-7).

An array of hanger node numbers (IHGRNODE) is read and written for each hanger in the input file and is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

The neutral file reads and writes an 11-element array (HGRDAT) for each hanger in the input file. The HGRDAT array is dimensioned (11,N5). Each hanger in the model requires two lines in the neutral file. Use the FORTRAN format (2X, 6G13.6).

1. Hanger stiffness
2. Allowable load variation
3. Rigid support displacement criteria
4. Allowed space for hanger
5. Cold load #1 (theoretical)
6. Hot load #1 (initialize to 0.0)
7. User defined operating load f/ variable springs (init to 0.0)
8. Maximum allowed travel limit
9. Multiple load case design option
10. Hanger hardware weight
11. Constant effort support load (CEFF)

Use the FORTRAN format (7X, I5, 1X, A100) to write the values of the following two items on the next two lines of the neutral file. These two items follow the hanger specifications above and the neutral file repeats the lines for each hanger defined in the Hangers Auxiliary data in the piping input.

1. Hanger Tag
2. Hanger GUID

 **NOTE** The value at the beginning of each string indicates the length of the string. If the string is blank, the value is 0.

A four-element array (IHGRFREE) is read/written for each hanger in the input file. The IHGRFREE array is dimensioned (4,N5). Each hanger in the file requires one line in the neutral file.

Use FORTRAN format (2X, 6I13).

1. Anchor node to be freed (#1)
2. Anchor node to be freed (#2)
3. d.o.f. type for #1 (1-free Y, 2-free XY, 3-free ZY, 4-free X, Y, Z, 5-free all)
4. d.o.f. type for #2

An array (IHGRNUM) lists the number of hangers at this location for each hanger in the input file. There is one entry here for every hanger in the file. The IHGRNUM array is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

An array (IHGRTABLE) lists the hanger table numbers for each hanger in the input file. There is one entry here for every hanger in the file. The IHGRTABLE is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

An array of flags (IHGRSHORT) indicates if short range springs can be used at each hanger location. The IHGRSHORT array is dimensioned (N5). There are seven lines in the neutral file for this data. Use the FORTRAN format (2X, 6I13).

- 0 = cannot use short range springs
- 1 = can use short range springs

An array of connecting node numbers (IHGRCN) is available for each hanger. The IHGRCN array is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

Execution Options

Defines the execution options used by CAESAR II. Use the FORTRAN format (2X, 4I13, G13.6, I13). This requires three lines in the neutral file. These values are:

- Print forces on rigids and expansion joints 0=no, 1=yes
- Print alphas & pipe props. during error checking 0=no, 1=yes
- Activate Bourdon Pressure Effects 0, 1, or 2
- Activate Branch Error and Coordinate Prompts 0=no, 1=yes
- Thermal Bowing Delta Temperature degrees
- Use Liberal Stress Allowable 0=no, 1=yes

For the following data, use the FORTRAN format: (2X, I13, 2G13.6, 3I13):

- Uniform Load Input in g's 0=no, 1=yes
- Stress Stiffening due to Pressure 0, 1, 2
- Ambient Temperature (If not 70.00 deg F) degrees
- FRP Expansion * 1,000,000 len/len/deg
- Optimizer 0-Both, 1-CuthillMcKee, 2-Collins
- Next Node Selection 0-Decreasing, 1-Increasing

For the following data, use the FORTRAN format (2X, 4I13, G13.6, I13):

- Final Ordering 0-Reversed, 1-Not Reversed
- Collins Ordering 0-Band, 1-No. of Coefficients
- Degree Determination 0-Connections, 1-Band
- User Control 0-None, 1-Allow User Re-Looping
- FRP Shear ratio

- Laminate type

For the following data, use the FORTRAN format (2X, I13):

- North arrow

Units Conversion Data

NOTE Where the FORTRAN format is (2X, 6G13.6), you can change the format to (2X, 6(1p,E13.6)) to write the value with seven significant digits.

#\$ UNITS

Defines both the conversion constants and the conversion labels. The conversion constants are all **REAL*4** values in FORTRAN format (2X, 6G13.6). This requires four lines in the neutral file. The character definitions for the labels are listed below.

CNVLEN

Defines the length conversion

CNVFOR

Defines the force conversion

CNVMAS

Defines the mass conversion

CNVMIN

Defines the moment (input) conversion

CNVMOU

Defines the moment (output) conversion

CNVSTR

Defines the stress conversion

CNVTSC

Defines the temperature conversion

CNVTOF

Defines the temperature offset

CNVPRE

Defines the Pressure conversion

CNVYM

Defines the Young's modulus conversion

CNVPDN

Defines the pipe density conversion

CNVIDN

Defines the insulation density conversion

CNVFDN

Defines the fluid density conversion

CNVTSF

Defines the translational stiffness conversion

CNVRSF

Defines the rotational stiffness conversion

CNVUNI

Defines the uniform load conversion

CNVGLD

Defines the G load conversion

CNVWND

Defines the wind load conversion

CNVELE

Defines the elevation conversion

CNVCLN

Defines the compound length conversion

CNVDIA

Defines the diameter conversion

CNVTHK

Wall thickness conversion

Next, enter the following labels for units, one per line, in the format given in the label descriptions. This requires 24 lines in the neutral file.

CCVNAME

Defines the name of the units used, such as English (CHARACTER*15)

CCVNOM

Sets On or Off, and tells PREPIP whether or not nominal diameters are allowed (CHARACTER* 3)

CCVLEN

Defines the length label (CHARACTER* 3)

CCVFOR

Defines the force label (CHARACTER* 3)

CCVMAS

Defines the mass label (CHARACTER* 3)

CCVMIN

Defines the moment (input) label (CHARACTER* 6)

CCVMOU

Defines the moment (output) label (CHARACTER* 6)

CCVSTR

Defines the stress label (CHARACTER*10)

CCVTSC

Defines the temperature label (CHARACTER* 1)

CCVTOF

Defines the temperature offset/label (CHARACTER* 1)

CCVPRE

Defines the pressure label (CHARACTER*10)

CCVYM

Defines Young's modulus label (CHARACTER*10)

CCVPDN

Defines the pipe density label (CHARACTER*10)

CCVIDN

Defines the insulation density label (CHARACTER*10)

CCVFDN

Defines the fluid density label (CHARACTER*10)

CCVTSF

Defines the translational stiffness label (CHARACTER* 7)

CCVRSF

Defines the rotational stiffness label (CHARACTER*10)

CCVUNI

Defines the uniform load label (CHARACTER* 7)

CCVGLD

Defines the gravitational load label (CHARACTER* 3)

CCVWND

Defines the wind load label (CHARACTER*10)

CCVELE

Defines the elevation label (CHARACTER* 3)

CCVCLN

Defines the compound length label (CHARACTER* 3)

CCVDIA

Defines the diameter label (CHARACTER* 3)

CCVTHK

Defines the wall thickness label (CHARACTER* 3)

Nodal Coordinate Data

#\$ COORDS

Specifies the X, Y, Z global coordinates of the starting node point of each discontinuous piping segment. The data in this section of the neutral file is optional; it may not exist. The existence of this data depends on user preference and the particular job. The data is defined below.

NXYZ

Defines how many sets of coordinates follow. Use FORTRAN format (2X, I13).

INODE, XCORD, YCORD, ZCORD

Consists of four values in a line and is repeated NXYZ times.

Use FORTRAN format (2X, I13, 3F13.4) to define a node number and the X, Y, Z global coordinates.

 **NOTE** This section only exists in Versions 3.22 and later.

Neutral File Generator

Generates the CAESAR II neutral file (.cii), which you use to send data to or from the standard CAESAR II input file to another interface or a third-party application.

Conversion Type

Specifies the type of file to convert using the Neutral File Generator. Convert a neutral file (.cii) to an CAESAR II input file (._a) or vice-versa.

Next, browse and select or type the name of the neutral file to be converted and select **Convert**. The software prompts you when the conversion completes. Select **OK** and indicate whether to perform another file conversion. When you are done converting files, select **No** and the software exits the dialog.

Neutral File Name to Be Converted

Browse and select or type the name of the file to be converted, based on the **Conversion Type** you selected.

CAESAR II indicates the name of the file and the directory. The software defaults this path to the same location as the file you selected for conversion.

Next, browse and select or type the name of the converted file and select **Convert**. The software prompts you when the conversion completes. Select **OK** and indicate whether you want to perform another file conversion. When you are done converting files, select **No** and the software exits the dialog.

File Name to Be Created

Indicates the name of the file the generator creates, which is the same name as the selected file for conversion, but with an extension that reflects the **Conversion Type** selected (i.e., `._a` for a CAESAR II input file and `.cii` for a neutral file).

Copyright

Copyright© Hexagon AB and/or its subsidiaries and affiliates. All rights reserved.